

Django Template Inheritance

Connecting Main Project Templates with App Templates to avoid code repetition.

🎯 1. Topic Overview & Goal

Is lecture ka main aim hai **Code Repetition (DRY Principle)** ko rokna. Har nayi HTML file mein `<head>`, `<html>` aur common UI (jaise header/footer) baar-baar likhne ke bajaye, hum ek **Base Template** banayenge aur usko baki Apps ke templates ke sath connect karenge.

💡 **Core Concept:** Project ke main template ko configure karke ek baar setup karna, aur phir us setup ko apni baaki sabhi applications (jaise Blog, Shop) mein **Reuse** karna.

🧩 2. Important Template Tags

🔗 `{% extends %}`

Yeh tag kisi child template mein sabse upar likha jata hai. Iska kaam base file ke structure ko **inherit (copy)** karna hota hai.

📦 `{% block %}` & `{% endblock %}`

Yeh base file mein **Dynamic Spaces (Empty containers)** create karte hain, jahan child files apna specific content daal sakti hain.

🔧 3. Step 1: Base Template Setup (base.html)

Sabse pehle main project ke `templates/base.html` mein jayenge aur usey dynamic banayenge. Jo cheezein common hain (jaise `<hr>` tag ya "Welcome" message) unhe waise hi chhod denge.

Dynamic Title & Content Banane ka Code:

```
<html> <head> <title> {% block title %} {% endblock title %} | My Django Project </title> </head> <body> <h2>Welcome to My Project</h2> <hr> <!-- Yeha par child apps apna content bhejengi --> {% block content %} {% endblock content %} </body> </html>
```

- `{% block title %}` : Yahan baaki apps apna custom page title bhejengi.
- `{% block content %}` : Yahan apps apna main HTML body content bhejengi.

4. Step 2: App Template ko Connect Karna

Ab hum apni App ke HTML files (e.g., `blog/post_list.html` ya `shop/product_list.html`) mein jayenge. Wahan se saara purana faltu boilerplate HTML (head, body) **delete** kar denge.

App Template (Child File) ka Code:

```
{% extends 'base.html' %} {% block title %} Blog Posts {% endblock title %}
{% block content %} <h1>Here is the Latest Blog Post</h1> <p>List of posts will
appear here...</p> {% endblock content %}
```

Fayda: Jaise hi `blog_post.html` render hoga, Django automatically `base.html` ko load karega aur Title ki jagah "*Blog Posts*" aur Content ki jagah hamara `<h1>` aur `<p>` insert kar dega.

5. Common Error: TemplateDoesNotExist

 **Problem:** Agar page load karne par `TemplateDoesNotExist` (`base.html`) error aaye, iska matlab Django ko aapka `templates` folder nahi mil raha hai.

 **Solution (Folder Placement):** Aapne galti se `templates` folder ko *Project Config Folder* (jisme `settings.py` hota hai) ke andar bana diya hoga. Use wahan se utha kar **Root Directory** (base directory) mein move karein. `templates` folder aapke baaki apps (Blog, Shop) aur `manage.py` ke **parallel** hona chahiye. Iske baad server ko refresh karein.

6. Quick Revision / Key Takeaways

- **DRY Principle:** HTML code baar-baar repeat na karein. Ek `base.html` banayein.
- **Dynamic Blocks:** `{% block name %}` aur `{% endblock %}` ke beech ka content override kiya ja sakta hai.
- **Extending:** Child template ki sabse **pehli line** hamesha `{% extends 'base.html' %}` honi chahiye.
- **Static vs Dynamic:** Base template ka jo hissa block ke bahar likha hota hai (jaise Navbars ya Footer), wo sabhi pages par *Static* (same) rehta hai.
- **Directory:** `templates` folder ko hamesha **Root** (`BASE_DIR`) mein rakhein.